

SOCIAL AND ECONOMIC IMPACTS OF THE FLOOD IN TUTICORIN
MAJOR PROJECT 2024-2025

Work done by

PANIMAYA PREETHI S

(22BBA529)

Under the Co-ordination of Mr

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ASSISTANT PROFESSOR

A Report Submitted in partial fulfillment of the requirements

for the award of the degree of

Bachelor of Business Administration



PSG COLLEGE OF ARTS & SCIENCE

An Autonomous College Affiliated to Bharathiar University

Accredited with A⁺⁺ Grade by NAAC(4th CYCLE)

College with Potential for Excellence (Status awarded by UGC)

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An ISO 9001:2015 Certified Institution

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Department of Management Sciences

Bachelor of Business Administration

CERTIFICATE

This is to certify that project report entitled “Social and Economic Impacts of the Flood in Tuticorin” is a bonafide record of original research work done by **Panimaya Preethi S(22BBA529)** submitted in the partial fulfillment for the award of the degree of Bachelor of Business Administration.

Faculty Coordinator

Head of the Department

Viva voce held on _____

Examiners 1. _____

2. _____

DECLARATION

I ,Ms.Panimaya Preethi S(22BBA529) hereby declare that this Major Project entitled “**Social And Economic Impacts Of The Flood In Tuticorin**” is an original work done by me ,under the co-ordination of Dr. K.M DEVARAJ., MBA.,Ph.D during my period of study 2024 – 2025.

DATE:

PLACE:

SIGNATURE OF THE STUDENT

ACKNOWLEDGMENT

It is great pleasure, deep satisfaction and gratitude that I acknowledge the contributors of the array of the individual towards the successful completion of the Major project.

At this outlet, I wish to extend my heart-felt sincere thanks to **Dr.D.Brindha M.Sc, MPhil, PhD, Principal**, PSG College of Arts & Science for giving me an opportunity to do this project.

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It is a great honor to thank Head of the Department, **Dr.KAVITHA RAMU MBA,MPhil, PGDHRM, PhD**, PSG College of Arts & Science for her encouragement during the major project.

I would like to place on record the sincere and earnest guidance extended to me, by my faculty co-ordinator , **Dr. K.M DEVARAJ, MBA.,PhD Assistant Professor**, for his guidance and encouragement provided, which was a great support for the completion of this major project.

PANIMAYA PREETHI S

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SOCIAL AND ECONOMIC IMPACT OF THE FLOOD IN TUTICORIN

CHAPTER - I

1.1 INTRODUCTION OF THE STUDY:

Tuticorin, a coastal town in Tamil Nadu, India, is susceptible to flooding due to its unique geographical and climatic conditions. The region's proximity to the Arabian Sea, coupled with monsoonal rainfall patterns, makes it particularly vulnerable to heavy rainfall and storm surges. The socio-economic landscape of Tuticorin, characterized by its reliance on fishing, shipping, and agriculture, is significantly impacted by flooding events.

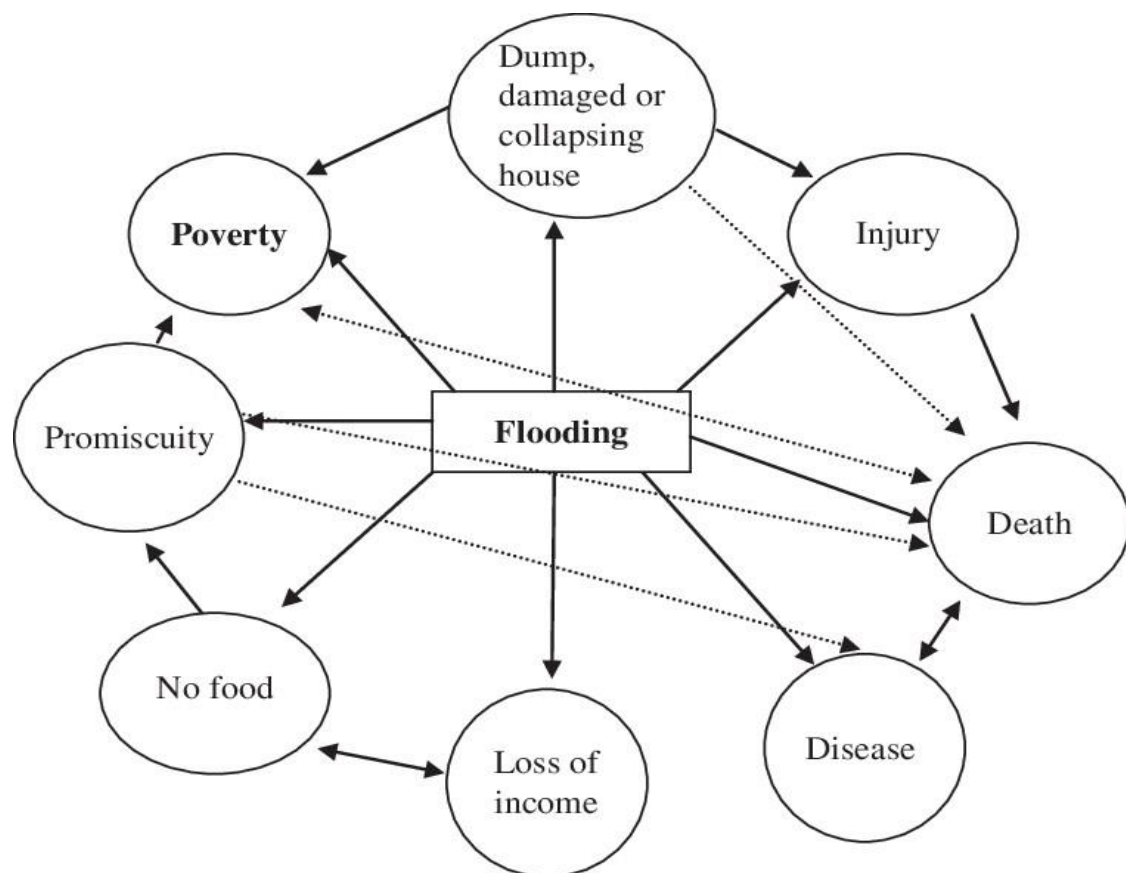
This study aims to assess the multifaceted impacts of floods in Tuticorin, focusing on the environmental, economic, and social dimensions. By examining past flood events, the research will analyse changes in land use, alterations in local ecosystems, and the subsequent effects on livelihoods. Additionally, it will explore community resilience and the effectiveness of existing flood management strategies.

Through a combination of quantitative data analysis and qualitative assessments, this study seeks to provide insights that can inform policy-making, enhance disaster preparedness, and promote sustainable development in the region. Understanding the implications of flooding in Tuticorin is crucial for mitigating risks and fostering a more resilient community in the face of climate change.

1.1.2 FLOODING:

The flood in Tuticorin had devastating social, economic, and environmental impacts. Many families were displaced, leading to temporary shelters and a breakdown of community networks. Health risks surged as stagnant waters facilitated the spread of waterborne diseases, while mental health issues arose from the trauma experienced. Educational institutions faced closures, disrupting the learning process for countless children

Economically, local businesses suffered severe losses due to damage and reduced foot traffic, while agriculture took a hit with flooded farmlands and destroyed crops. Infrastructure damage hindered mobility and access to essential services, compounding the crisis. Environmental degradation occurred as ecosystems were disrupted, affecting local flora and fauna. Contaminated water sources raised concerns about long-term health impacts. Recovery efforts strained local resources, necessitating external aid and support. Overall, the flood's aftermath highlighted the urgent need for better preparedness and resilience strategies in the face of such disasters.



1.1.3 HISTORY OF FLOOD IN TUTICORIN

Thoothukudi, located along the southeastern coast of India, has experienced several significant flooding events over the last century, largely influenced by its coastal geography and climatic conditions.

Early 20th Century

In the early 1900s, Thoothukudi faced sporadic flooding, primarily due to heavy monsoonal rains. However, records from this period are sparse, making it difficult to assess the full impact of these events.

1950s-1970s

The floods in this period were marked by the devastating monsoon seasons. In 1966, heavy rainfall resulted in severe flooding that disrupted local agriculture and fisheries, which are vital to the region's economy. This event highlighted the vulnerability of Thoothukudi's infrastructure and the need for improved flood management systems.

1980s-1990s

The late 20th century saw a few notable floods, including a significant event in 1988, when the area experienced unprecedented rainfall. The flooding caused extensive damage to homes, agriculture, and local businesses, leading to displacement and economic hardship for many residents.

2000s

In 2005, Thoothukudi faced severe flooding due to the combined effects of heavy monsoon rains and high tides. This event prompted the government to implement some flood mitigation measures, including improvements to drainage systems. However, the rapid urbanization of the town continued to exacerbate flooding risks.

Recent Events

The floods of 2015 were particularly impactful, with heavy rainfall leading to widespread inundation. The local administration faced challenges in managing the floodwaters, resulting in significant damage to infrastructure and livelihoods.

In 2021, Thoothukudi experienced additional flooding during the Northeast monsoon, prompting emergency responses and highlighting ongoing vulnerabilities in flood management.

In 2023, Tuticorin experienced significant flooding due to intense monsoon rains, exacerbated by high sea levels. Heavy downpours led to waterlogging in many areas, disrupting transportation and damaging homes and infrastructure. Local rivers overflowed, affecting agricultural fields and displacing several families.

1.2 FOCUS OF THE STUDY

This study aims to investigate the social and economic impacts of flooding in Tuticorin, with a particular emphasis on the following key areas:

1. **Community Displacement and Vulnerability:** Assessing how floods affect the displacement of communities, particularly marginalized groups, and exploring their access to resources and recovery support.
2. **Economic Losses:** Analysing the direct and indirect economic impacts on local industries, including fishing, agriculture, and small businesses, to quantify financial losses and disruptions.
3. **Health and Well-being:** Examining the effects of flooding on public health, including waterborne diseases and mental health challenges, as well as access to healthcare services post-flood.
4. **Infrastructure and Services:** Evaluating the impact of floods on critical infrastructure— such as roads, schools, and sanitation facilities—and how this affects daily life and long-term development
5. **Government response:** Investigation how government respond to the situation and actions taken by the government and satisfaction of the people towards it.

This comprehensive focus aims to provide a nuanced understanding of how floods shape social dynamics and economic conditions in Tuticorin, informing future disaster management and policy-making efforts.

1.3 NEED FOR THE STUDY:

Increased Frequency of Flood Events: With climate change contributing to more frequent and intense flooding, it is essential to understand the evolving social and economic impacts on communities like Tuticorin.

Vulnerability of Local Economies: Tuticorin's economy is heavily reliant on sectors such as fishing and agriculture, which are particularly susceptible to flooding. Understanding these impacts can help develop targeted support measures.

Displacement and Social Equity: Floods often disproportionately affect marginalized communities. This study can highlight disparities in vulnerability and recovery, informing more equitable disaster response and recovery strategies.

Policy Development: Insights from this study can aid local and state authorities in designing effective flood management policies, improving infrastructure resilience, and enhancing community preparedness for future events.

Evaluation of Response Effectiveness: Understanding the effectiveness of government actions during past flood events is essential for assessing strengths and weaknesses in disaster response strategies.

Community Communication: Analysing government communication strategies during floods can reveal how effectively information was disseminated to the public, impacting community preparedness and response.

Policy Improvement: A study of past government actions can inform policy adjustments, helping to create more robust frameworks for disaster management and resilience-building in vulnerable areas.

1.4 OBJECTIVES:

- To know the major affected areas
- To find the impacts of the flood
- To analyse the actions taken by the government.

The goal is to pinpoint specific geographic regions within Tuticorin that experienced the most severe impacts from flooding. Understanding these areas will help in targeted relief efforts, policy formulation, and future preparedness.

This objective aims to understand the various consequences of flooding on both the social and economic fabric of the affected areas in Tuticorin. By thoroughly examining these impacts, we can develop better strategies for mitigation, recovery, and future preparedness.

This objective aims to evaluate the effectiveness and efficiency of government responses to flooding in Tuticorin. By analysing these actions, we can identify strengths, weaknesses, and areas for improvement in disaster management strategies.

1.5 LIMITATIONS OF THE STUDY:

- Temporal Limitations

Short-term Focus: A focus on immediate post-flood impacts might neglect longer-term consequences, such as ongoing economic recovery or social adaptation processes.

Evolving Conditions: Economic and social conditions can change rapidly, making it challenging to capture the full impact over time.

- Economic Complexity

Difficulty in Quantifying Losses: Estimating economic losses, especially in informal sectors or small businesses, can be challenging and may result in underreporting.

Interconnected Economic Factors: Other economic factors (e.g., market fluctuations) unrelated to flooding may influence recovery outcomes.

- Political and Social Sensitivity

Community Trust Issues: Distrust in external researchers may lead to reluctance in sharing information, particularly in vulnerable populations.

Political Influences: Government narratives or biases may influence data availability and reporting, impacting the study's findings.

- External Factors

Impact of Climate Change: Ongoing climate changes can complicate assessment of flood impacts and recovery efforts over time.

Compounding Disasters: Other concurrent disasters (e.g., pandemics, economic downturns) may influence the social and economic landscape complicating the analysis.

1.6 RESEARCH GAP:

There are wide number of studies has been already done on the flood impact ,but there is lack of research being done in flood impact in Tuticorin.

While existing studies may focus on immediate impacts of flooding in Tuticorin, there is a lack of longitudinal research that examines the long-term social and economic recovery processes. Specifically, the gap lies in understanding how different communities adapt to post-flood conditions over time, including variations in recovery rates across socioeconomic strata.

1.7 SCOPE OF THE STUDY:

- Impact assessment: Evaluate the socio-economic, environmental and health impacts of the flood on affected areas and population
- Policy analysis: Analyse government policies, regulations and institutional frameworks related to flood risk management and development of the risk reduction.
- Risk assessment and Management: Development models and rules for assessing flood risk, implementing risk reduction measures , and improving emergency response and recovery plans.
- There are research areas offers opportunities to contribute to understanding, mitigating, and managing the challenges faced by flood-affected areas and promoting sustainable development and resilience.

1.8 STATEMENT OF THE PROBLEM:

- ✚ **Economic impact** : Flood can disrupt livelihood , agriculture , business, and transportation system, causing economic hardship for individual and communities.
- ✚ **Loss of lives:** Flood can result in the tragic loss of lives , impacting families and communities.
- ✚ **Property damage:** Floodwater can cause significant damage to homes , buildings infrastructure , and personal belongings leading to financial strain and displacement.
- ✚ **Water damage:** Flooding causes extensive damage to homes , buildings , infrastructure and agriculture land. This damage can lead to financial losses and displacement of resident.

1.9 HYPOTHESIS:

Floods can lead to severe disruptions in communities, impacting both the social fabric and economic stability. In Tuticorin, a coastal city in Tamil Nadu, floods can affect infrastructure, livelihoods, health, and community cohesion.

✚ Null Hypothesis:

There is no association between damages experienced by the respondents and their satisfaction level from the government's compensation.

✚ Alternative Hypothesis:

There is association between damages experienced by the respondents and their satisfaction level from the government's compensation.

CHAPTER II

LITERATURE REVIEW:

A literature review is a detailed analysis and summary of the existing research on the particular topic, designed to provide a comprehensive overview of the current knowledge, identify gaps and suggest areas for future study. Previous studies have approached flood from various angles, includes migration, impact, environmental loss, etc. Despite the rich body of literature, gaps remain in people satisfaction towards government actions. This review will not only address these gaps but also provide a comprehensive overview of the current state of research, offering insights into the complexities of flood and suggesting directions for future inquiry.

1. Singh, R. (2015)

Literature review for the study about Historical Analysis of Flood Events in South India, provides a comprehensive examination of flood occurrences in the region, focusing on the factors contributing to their frequency and intensity. The study is significant in contextualizing the broader impacts of flooding on both societal structures and economic stability in South India, with implications for disaster management and policy formulation.

2. Kumar.A & Rao.S (2016)

This study identifies various factors contributing to community vulnerability, including socioeconomic status, access to resources, and infrastructural weaknesses. Their analysis reveals that poorer communities are disproportionately affected by flooding, suffering greater losses and slower recovery times. Kumar and Rao highlight the interplay between local governance, community engagement, and the effectiveness of flood preparedness measures.

3. Sharma, P. (2017)

This study identifies several key impacts of flooding on agricultural systems, including crop loss, soil degradation, and changes in water quality. The study reveals that floods not only lead to immediate crop damage but also have long-term effects on soil fertility and agricultural sustainability. Additionally, the author notes that smallholder farmers are particularly at risk, as they often lack the resources to recover effectively from flood damage

4. Menon, S. et al. (2018)

This study about a flood identifies a range of health risks associated with flooding, including the outbreak of waterborne diseases, such as cholera and typhoid, as well as vector-borne diseases like dengue and malaria. Menon et al. emphasize that the disruption of sanitation and access to clean drinking water during floods exacerbates these health risks. Additionally, the study highlights the psychological impacts of flooding, including increased anxiety and post-traumatic stress disorder (PTSD) among affected populations.

5. Gupta and Verma's (2019)

This study identify a range of health risks associated with flooding, including the outbreak of waterborne diseases, such as cholera and typhoid, as well as vector-borne diseases like dengue and malaria. Menon et al. emphasize that the disruption of sanitation and access to clean drinking water during floods exacerbates these health risks. Additionally, the study highlights the psychological impacts of flooding, including increased anxiety and post- traumatic stress disorder (PTSD) among affected populations.

6. Chatterjee, R. (2020)

The study identifies significant patterns of displacement linked to flooding, highlighting that both temporary and permanent migration often occurs in the aftermath of severe flood events. Chatterjee reports that families from economically disadvantaged backgrounds are more likely to be displaced, leading to increased vulnerabilities and social fragmentation. The research also notes the challenges faced by displaced populations in accessing basic services, housing, and employment opportunities.

7. Rani. L and Subramanian. K(2021)

The authors identify a range of response strategies implemented by both government agencies and NGOs, highlighting collaborative efforts that aim to enhance flood preparedness and recovery. Rani and Subramanian emphasize the importance of timely communication, resource allocation, and community engagement in the success of these initiatives. They found that while government responses often focus on infrastructure and immediate relief, NGOs frequently address longer-term recovery and community needs, such as psychosocial support and livelihood restoration.

8. Joseph. A (2023)

Joseph identifies several critical components of community resilience, including social cohesion, local knowledge, infrastructure preparedness, and access to resources. The study reveals that communities with strong social networks and active participation in disaster preparedness are better equipped to respond to floods and recover more effectively. Additionally, Joseph highlights the importance of integrating traditional knowledge and practices into modern resilience-building strategies, as these can enhance community-specific responses to flooding.

9. Das, M. & Reddy, V. (2023)

This investigation identifies key areas of infrastructure that are most vulnerable to flood damage, including transportation networks, water supply systems, and sanitation facilities. Their findings indicate that inadequate drainage systems and poor urban planning exacerbate the impacts of flooding, leading to significant economic losses and disruptions to daily life. Das and Reddy emphasize that the recovery of infrastructure is not only about physical repair but also about enhancing resilience to future flood events.

10. Patel, S. (2023)

Patel identifies several key vulnerabilities that emerge in the aftermath of flooding. The study reveals that women often bear a disproportionate burden due to existing socio-economic inequalities, such as limited access to resources, decision-making power, and information. Women are more likely to experience negative health outcomes, increased caregiving responsibilities, and economic instability following flood events. Patel emphasizes that these gendered vulnerabilities are compounded by cultural norms and expectations that restrict women's mobility and access to aid.

11. Iyer, N. & Narayan, P. (2023)

Iyer and Narayan's "Economic Resilience and Recovery Post-Flood in Coastal Regions" contributes significantly to the understanding of how communities can enhance their economic resilience in the face of flooding. By highlighting effective recovery strategies and the importance of local engagement, this research offers critical recommendations for policymakers, disaster management professionals, and community leaders seeking to build sustainable and resilient coastal economies.

12. Rao (2023)

Rao identifies several technological innovations that have significantly impacted flood management, including remote sensing, geographic information systems (GIS), and predictive modeling¹. The study reveals that these tools enable better forecasting of flood events, more accurate mapping of affected areas, and enhanced decision-making processes during disasters. Rao also discusses the integration of community-based technologies, such as mobile applications for disaster reporting and real-time communication, which empower local populations to engage in flood management proactively.

13. Srinivasan, R. (2023)

The authors identify several key factors that influence economic resilience, including local governance, community engagement, and access to financial resources. Iyer and Narayan found that communities with strong local leadership and effective communication networks recover more quickly from floods. They also highlight the role of microfinance and insurance in facilitating recovery, noting that these financial tools can empower households and small businesses to bounce back more effectively after a disaster.

14. Kaur, D. (2024)

This study identifies several critical aspects of climate adaptation that are essential for effective flood management. The study reveals that successful integration involves not only technical solutions, such as improved drainage systems and flood barriers, but also policy reforms and community engagement. Kaur emphasizes the importance of using climate data and predictive modelling to inform decision-making, allowing for more effective resource allocation and risk assessment. The research also highlights case studies where adaptive measures have significantly reduced vulnerability and improved recovery outcomes.

15. Nambiar, S. (2024)

The study reveals that communities with high levels of engagement and participation in flood preparedness activities are significantly more resilient. Nambiar identifies several successful models of community engagement, including the formation of local flood response teams and participatory risk assessment workshops. The research highlights how these initiatives foster a sense of ownership and responsibility among community members, leading to better preparedness and faster recovery following flood events.

CHAPTER III

RESEARCH METHODOLOGY:

Research methodology is the structured, theoretical analysis of the approaches and techniques employed in a particular field of study. It serves as the backbone of research, providing a systematic framework for collecting, analysis, and interpreting data. Importantly, research methodology addresses the validity and reliability of the study, ethical considerations such as confidentiality and informed consent, and the theoretical framework guiding the research.

RESEARCH DESIGN:

Research design is the detailed plan for conducting a study. The design specifies the type of research, the research setting, the sampling methods, and the data collection techniques. It also includes considerations for controlling variables and ensuring validity and reliability. By establishing a clear and structured approach, research design ensures that the study is methodologically sound, enabling accurate and replicable results. It helps in minimizing biases, optimizing resources use, and enhancing the overall rigor and credibility of the research process.

POPULATION OF THE STUDY:

Population of the study refers to the entire group that the researcher wants to draw the conclusion about, it may be either finite or infinite. It is completely based on the area of the study and the respondents. The population of this study is people in the major flood affected areas in Tuticorin.

SAMPLING TECHNIQUE:

There are two types of sampling techniques which is:

1)PROBABILITY SAMPLING:

Probability sampling is a sampling method in which the researcher selects criteria and randomly selects members of the population. This selection parameter gives all members an equal opportunity to participate in the sample. There are 4 types in probability samplings, they are simple random sampling, cluster sampling, systematic sampling and stratified random sampling.

2)NON-PROBABILITY SAMPLING:

In non-probability sampling, researchers randomly select members for the study. This sampling method is not a fixed or predefined selection process. This ensures that all elements of the population have an equal opportunity to be included in the sample. There are 4 types of non-probability sampling which are convenience sampling, judgemental sampling, snowball sampling and quota sampling.

In this study **probability sampling** is employed under that convenience sampling method has been used. The research is conducted by collecting data only from the people in the major flood affected areas in Tuticorin.

SAMPLE SIZE:

The sample size of the study is 160 respondents. Due to limited time constraints of the study the sample size has been restricted to 160.

METHODS OF DATA COLLECTION:

The data has been collected under two methods. One is primary data collection method and another one is secondary data collection method. The primary data collection has been carried out through a well structured questionnaire and a proper discussion with the respective respondents. Then the secondary data collection has been carried out through previous research papers, articles, records, magazines, journals and also in books related to the study.

TOOLS USED FOR DATA COLLECTION:

The tools used for data collection in this study is a well structured questionnaire.

ANALYSIS TOOLS:

The statistical analysis tool employed for this study is CHI-SQUARE TEST.

chi-square test: taken into consideration of a dependent variable and a independent variable, the test is done through the chi-square formula.

$$\chi^2 = \sum E(O-E)^2$$

$$\chi^2 = \text{Ohi-squared}$$

$$O_i = \text{Observed value}$$

$$E_i = \text{Expected value}$$

The null hypothesis and alternative hypothesis are framed.

If the value of P is higher than the level of significance, null hypothesis is accepted and alternative is rejected.

If the value of P is lower than the value of significance, alternative hypothesis is accepted and null hypothesis is rejected.

DEGREE OF FREEDOM:

The term degree of freedom refers to the number of independent values or quantities that can vary in the analysis without violating any given constraints. It helps in determining the reliability and variability of statistical estimates, influencing the shapes of various statistical distributions used in hypothesis testing and confidence interval estimation. Degree of freedom is utilized to determine if a certain null hypothesis can be rejected based on the total number of variables and samples within the experiment.

LEVEL OF SIGNIFICANCE:

In this study, the level of significance is pre-set as 5%(0.05). If the p-value is less than or equal to the significance level, reject the null hypothesis. If the p-value is larger than the significance level, fail to reject the null hypothesis.

CHAPTER – IV

GENDER OF THE RESPONDENTS

The table – 1 classifies the respondents into male and female

TABLE - 1

S.NO	PARTICULARS	NO.OF RESPONDENTS	PERCENTAGE
1	Male	105	65.6
2	Female	55	34.4
	TOTAL	160	100

SOURCE: Primary Data

INTERPRETATION:

The table – 1 represents that (65.6%) of the respondents are male and (34.4%) of the respondents are female.

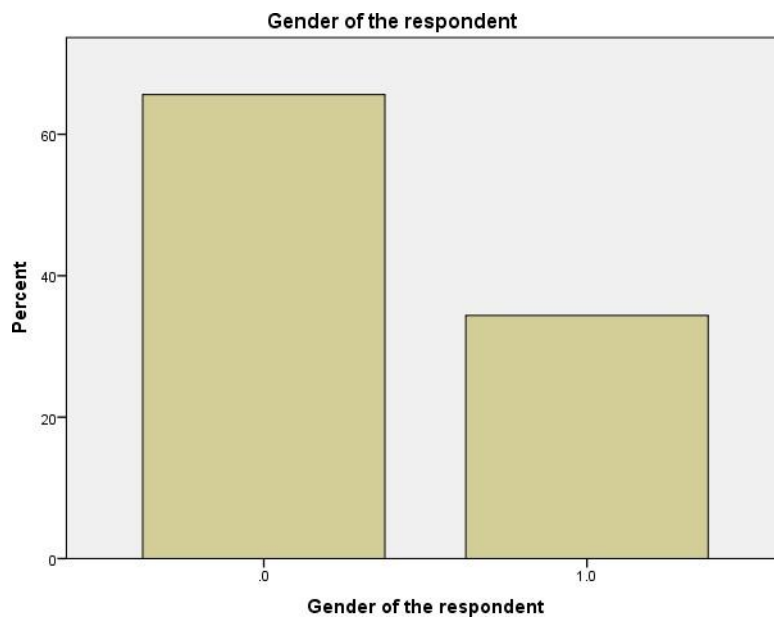


CHART-1

AGE OF THE RESPONDENT

TABLE - 2

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Below 20	25	15.6
2	20-29	91	56.9
3	30-39	32	20
4	Above 40	12	7.5
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-2 represents that (15.6%) of the respondents are under 20, (56.9%) of the respondents are 20-29 years old, (20%) of the respondents are 30-39 years old, (7.5%) of the respondents are above 40 years old.

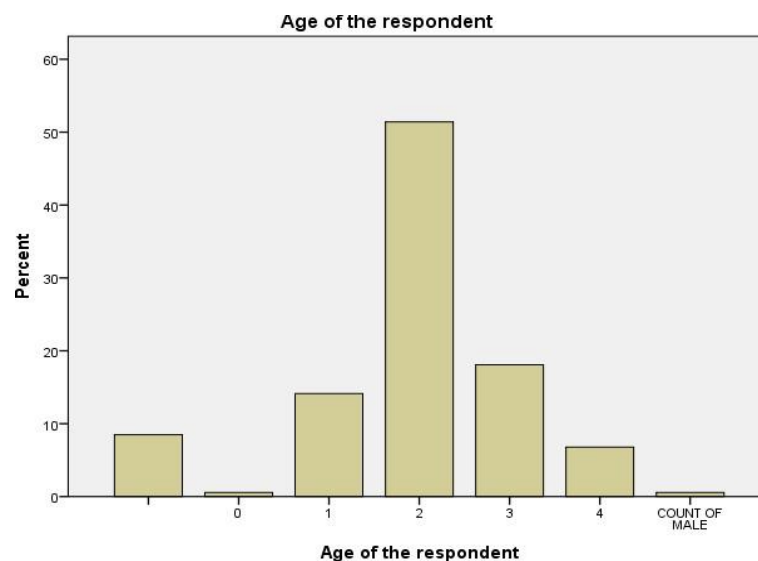


CHART-2

PLACE OF THE RESPONDENT

TABLE - 3

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Thiresh nagar	47	29.4
2	JJ nagar	42	26.3
3	Gandhi nagar	52	32.5
4	Muthaiya puram	19	11.9
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-3 represents that 29.4% of the respondents are from thiresh nagar, 26.3% off the respondents are from JJ nagar, 32.5% of the respondents are from Gandhi nagar, 11.9% of the respondents are from Muthaiya puram. So maximum no. of respondents are from thiresh nagar and minimum no. of respondents are from Muthaiya puram.

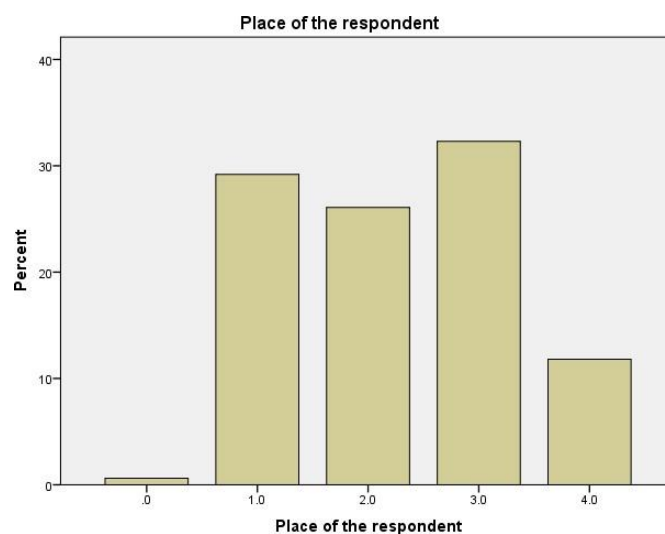


CHART-3

AFFECTED BY THE FLOOD

TABLE - 4

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Extremely	67	41.9
2	Significantly	62	38.8
3	Moderately	26	16.2
4	Slightly	5	3.1
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-4 represents that 41.9% of the respondents are extremely affected by the flood, 38.8% of the respondents are significantly affected by the flood, 16.2% of the respondents are moderately affected by the flood and 3.1% of the respondents are slightly affected by the flood.

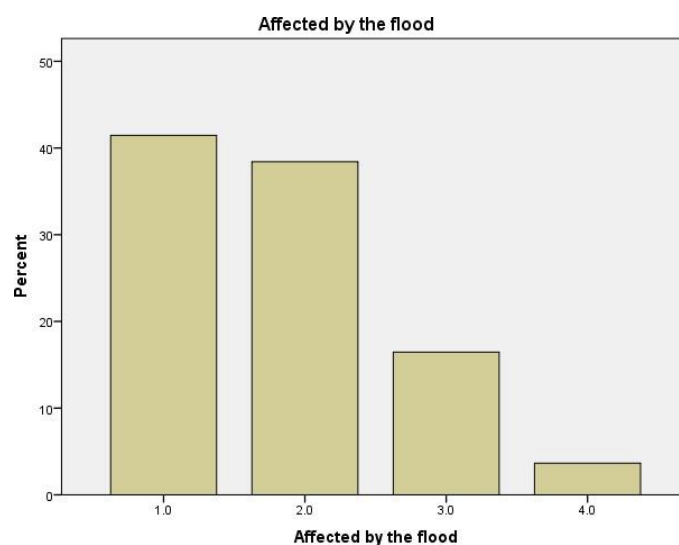


CHART-4

IMPACT DUE TO THE FLOOD
TABLE - 5

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Damage to the property	52	32.5
2	Loss of belongings	69	43.1
3	Health issues	24	15
4	Loss of income or livelihood	15	9.5
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-5 represents that 32.5% of the respondents experienced the damage to their property like house and vehicle, 43.1% of the respondents lose their belongings, 15% of the respondents faced health issues, 9.5% of the respondents affected by the loss of income or livelihood. As major no. of the respondents experienced loss of belongings major impact of the flood is considered as loss of belongings like vehicle, documents, things, etc.

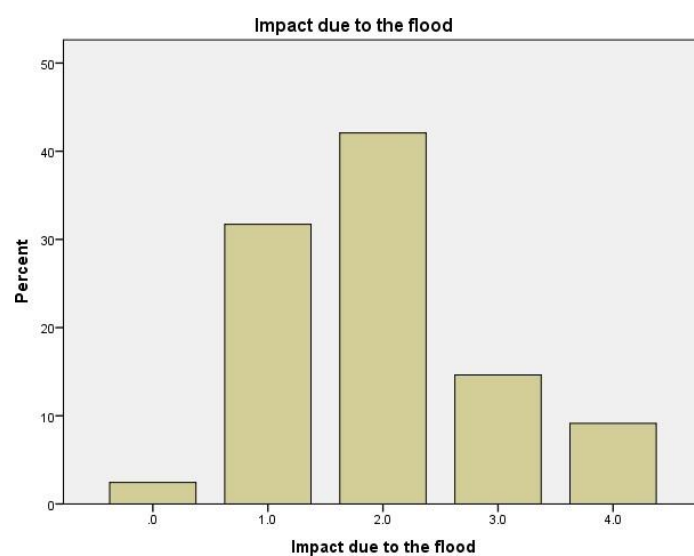


CHART-5

PEOPLE AFFECTED BY THE FLOOD

TABLE – 6

S.NO	PARTICULAR	NO.OF RESPONDENT	PERCENTAGE
1	Below 300	26	16.3
2	Around 500	60	37.5
3	Around 800	41	25.6
4	More than 1000	33	20.6
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-6 represents that 16.3% of the respondents consider around 300 people affected by the flood in their area, 37.5% of the respondents consider around 500 people affected by the flood in their area, 25.6% of the respondents consider around 800 people affected by the people in their area, 20.6% of the respondents consider more than 1000 people affected the flood in their area.

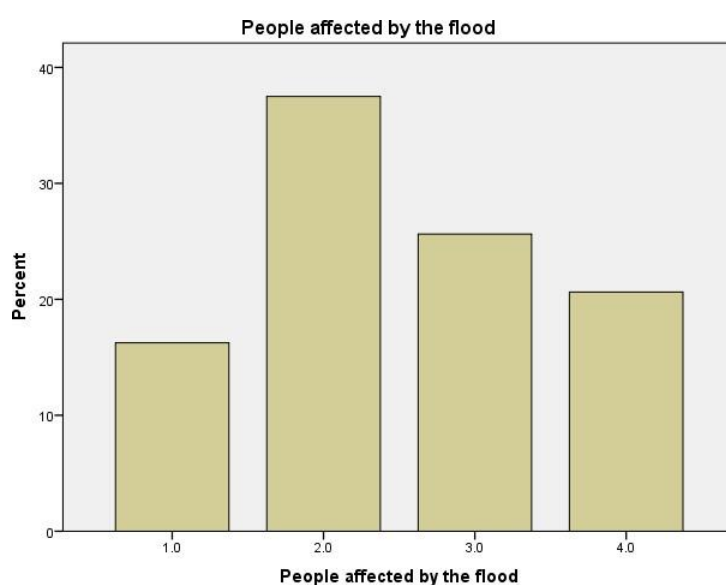


CHART-6

PERCENTAGE

PROPERTY DAMAGE EXPERIENCED DUE TO THE FLOOD

TABLE - 7

S.NO	PARTICULAR	NO.OF RESPONDENT	
1	House damage	40	25
2	Vehicle damage	69	43.1
3	Land	33	20.6
4	Agriculture	18	11.3
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION

The table-7 represents that 25% of the respondents affected by the house damage by the flood, 43.1% of the respondents affected by the vehicle damage by the flood, 20.6% of the respondents affected by the land damage by the flood, 11.3% of the respondents affected by the agriculture damage by the flood. So majority of the respondents faced vehicle damage.

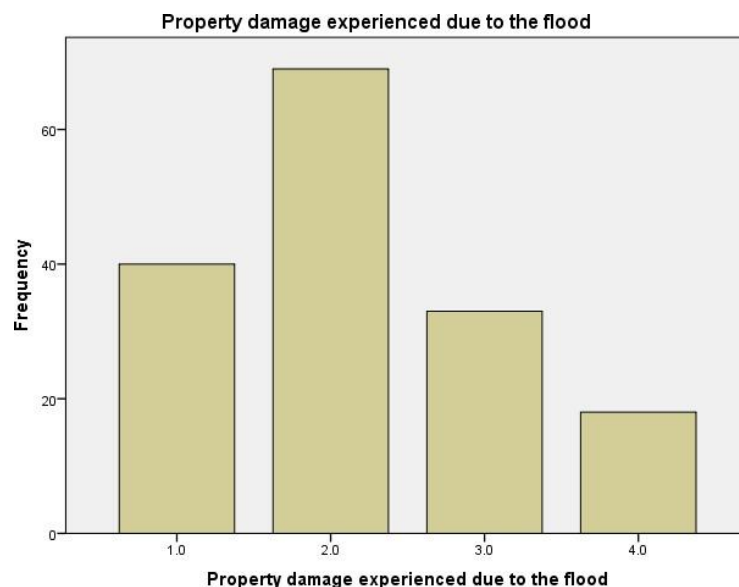


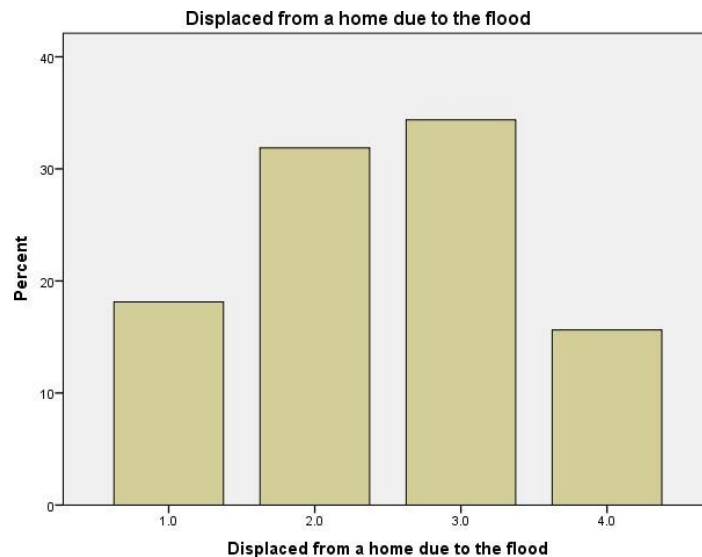
CHART-7

DISPLACED FROM A HOME DUE TO THE FLOOD**TABLE – 8**

S.NO	PARTICULAR	NO OF RESPONDENT	
1	One week	29	18.1
2	Two weeks	51	31.9
3	Three weeks	55	34.4
4	Not at all	25	15.6
	TOTAL	160	100

SOURCE: Primary data**INTERPRETATION:**

The table-8 represents 18.1% of the respondents displaced from home for one week, 31.9% of the respondents displaced from home for two weeks, 34.4% of the respondents displaced from home for three weeks, 15.6% of the respondents are not at all displaced from their home.

**CHART-8**

PERCENTAGE

DAYS ELECTRICITY WERE DOWN DURING THE FLOOD

TABLE - 9

S.NO	PARTICULAR	NO. OF RESPONDENT	
1	2-3 days	17	10.6
2	One week	45	28.1
3	Two weeks	72	45
4	Around a month	26	16.2
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-9 represents that 10.6% of the respondents experienced 2-3 days electricity down, 28.1% of the respondents experienced one week of electricity down, 45% of the respondents experienced two weeks of electricity down, 16.2% of the respondents experienced a month of electricity down. So majority of respondents affected by two weeks of electricity down.

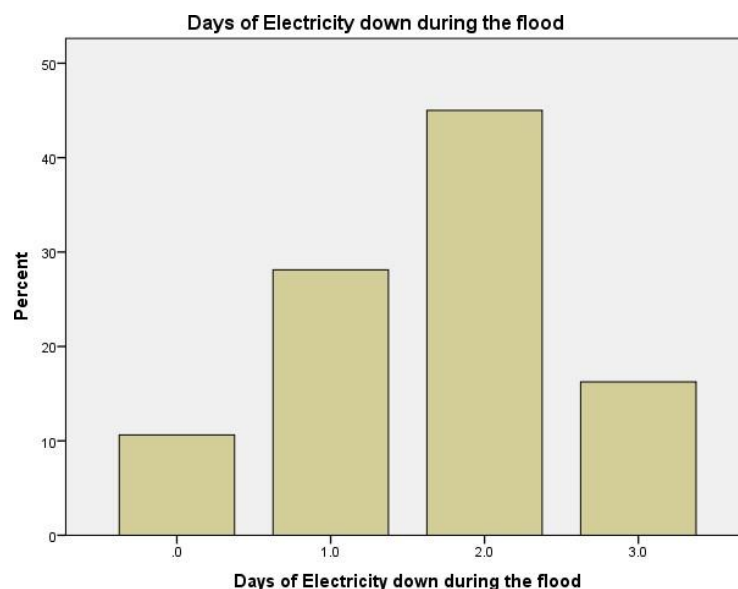


CHART-8

AFFECT ON HOUSEHOLD INCOME

TABLE – 10

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Reduced income	61	38.1
2	Loss of job	69	43.1
3	Business interruption	30	18.8
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-10 represents that 38.1% of the respondents affected by reduced income, 43.1% of the respondents affected by loss of job, 18.8% of the respondents affected by business interruption during the flood. So majority of respondents lose their job because of flood.

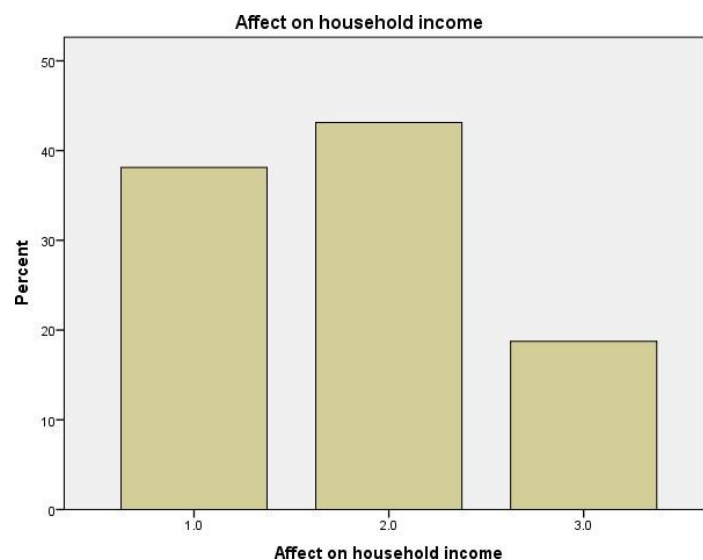


CHART-10

IMPACT ON DAILY LIFE ROUTINE

TABLE - 11

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Extremely	62	38.8
2	Significantly	87	54.3
3	Moderately	11	6.9
4	Slightly	0	0
	TOTAL	160	100

SOURCE: Primary data INTERPRETATION:

The table-11 represents that 38.8% of the respondent's daily life routine was extremely affected by the flood, 54.3% of the respondent's daily life routine was significantly affected by the flood, 6.9% of the respondent's daily life routine was moderately affected by the flood and there is no respondent who is not affected by the flood.

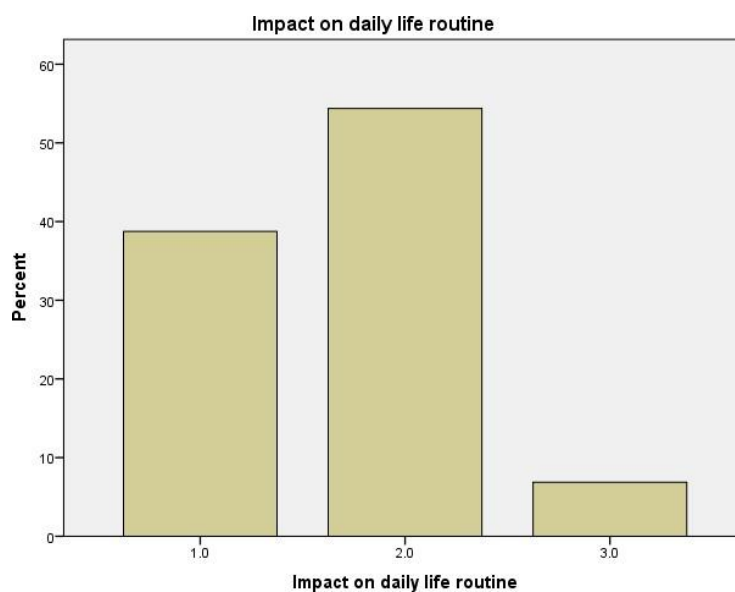


CHART-11

SUFFERED FOR FOOD DURING THE FLOOD

TABLE – 12

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Extremely	57	35.6
2	Significantly	80	50
3	Moderately	20	12.5
4	Slightly	3	1.9
	TOTAL	160	100

SOURCE: **Primary data**

INTERPRETATION:

The table-12 represents that 35.6% of the respondents are extremely suffered for food during the flood, 50% of the respondents are significantly suffered for food during the flood, 12.5% of the respondents are moderately suffered for food during the flood, 1.9% of the respondents are slightly suffered for food during the flood.

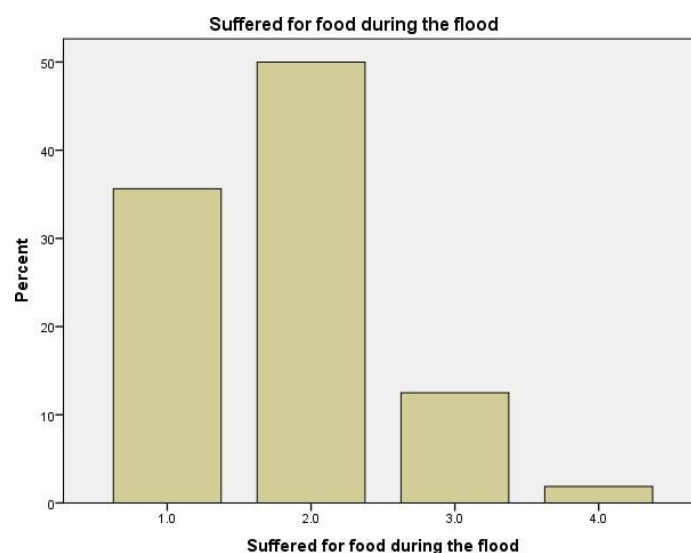


CHART-12

HEALTH ISSUES FACED DUE TO THE FLOOD

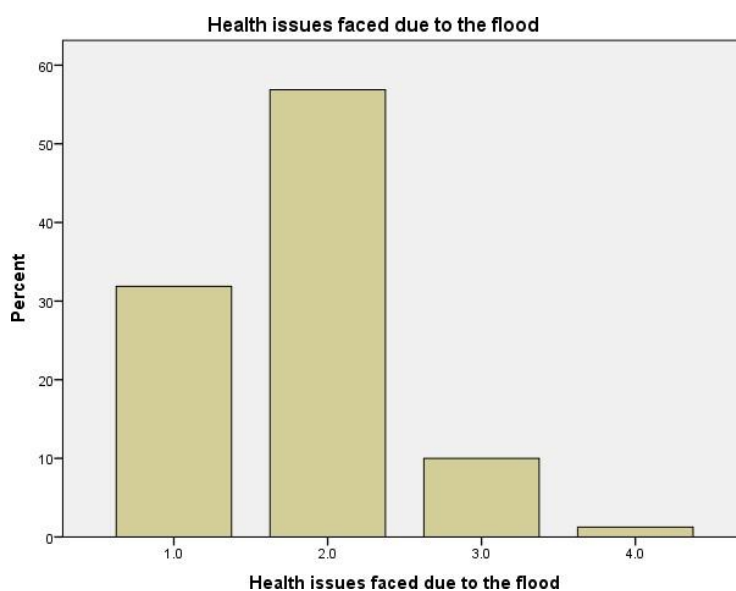
TABLE - 13

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Extremely	51	31.9
2	Significantly	91	56.9
3	Moderately	16	10
4	Slightly	2	1.2
		160	100

SOURCE: Primary data

INTERPRETATION:

The table-13 represents that 31.9% of the respondents are extremely suffered by health issues due to the flood, 56.9% of the respondents are significantly suffered by health issues due to the flood, 10% of the respondents are moderately suffered by health issues due to the flood, 1.2% of the respondents are slightly suffered by health issues due to the flood.



‘CHART-13

EMOTIONAL IMPACT DUE TO THE FLOOD

TABLE - 14

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Extremely	44	27.5
2	Significantly	85	53.1
3	Moderately	29	18.1
4	Slightly	2	1.3
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-14 represents that 27.5% of the respondents are extremely faced emotional impact of the flood, 53.1% of the respondents are significantly faced emotional impact of the flood, 18.1% of the respondents are moderately faced emotional impact of the flood, 1.3% of the respondents are slightly faced emotional impact of the flood.

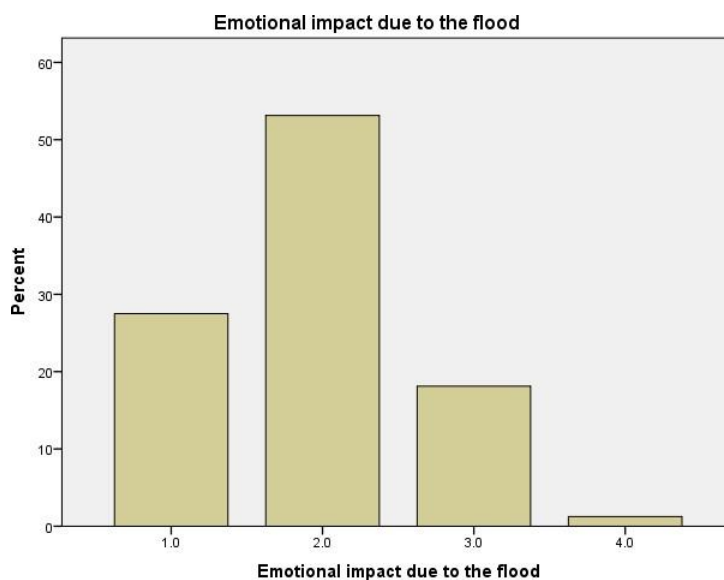


CHART-14

AFFECT ON SOCIAL INTERACTION

TABLE - 15

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Extremely	50	31.3
2	Significantly	81	50.6
3	Moderately	25	15.6
4	Slightly	4	2.5
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-15 represents that 31.3% of the respondents are extremely faced affect on social interaction, 50.6% of the respondents are significantly faced affect on social interaction, 15.6% of the respondents are moderately faced affect on social interaction, 2.5% of the respondents are slightly faced affect on social interaction.

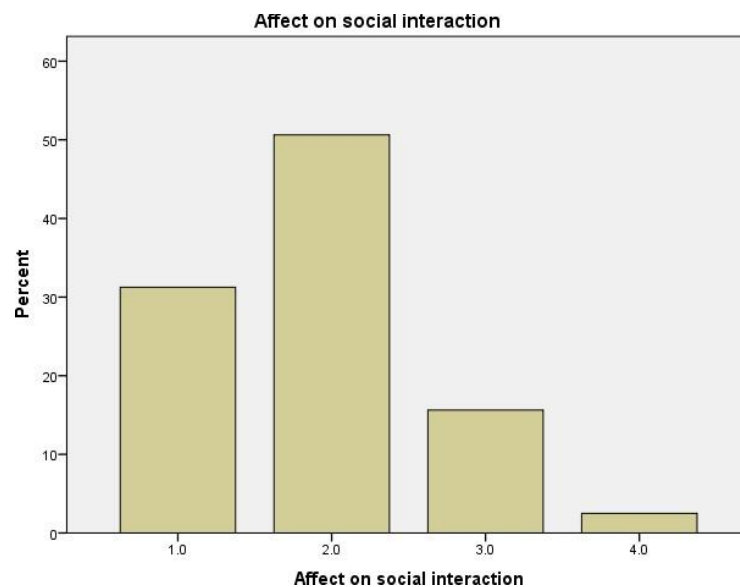


CHART-15

AFFECT ON CHILDREN EDUCATION

TABLE – 16

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Extremely	49	30.6
2	Significantly	80	50
3	Moderately	24	15
4	Slightly	7	4.4
	TOTAL	160	100

SOURCE: **Primary data**

INTERPRETATION:

The table-16 represents that 30.6% of the respondents are extremely faced affect on children education, 50% of the respondents are significantly faced affect on children education, 15% of the respondents are moderately faced affect on children education, 4.4% of the respondents are slightly faced affect on children education.

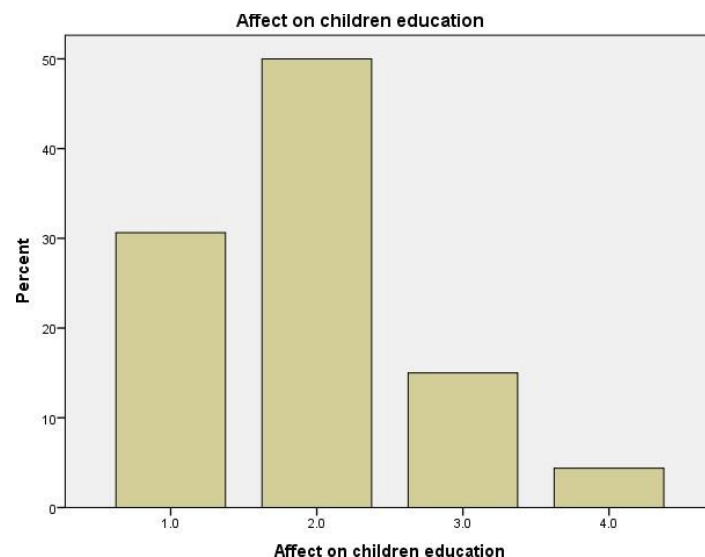


CHART-16

BUSINESS PERFORMANCE (BEFORE THE FLOOD)

TABLE - 17

S.NO	PARTICULARS	NO. OF RESPONDENT	PERCENTAGE
1	Excellent	38	23.8
2	Growing	89	55.6
3	Struggling	29	18.1
4	Loss	4	2.5
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION

The table-17 represents that 23.8% of the respondents experienced excellent business performance before the flood, 55.6% of the respondents experienced growing business performance before the flood, 18.1% of the respondents experienced struggling business performance before the flood, 2.5% of the respondents experienced loss in business performance before the flood.

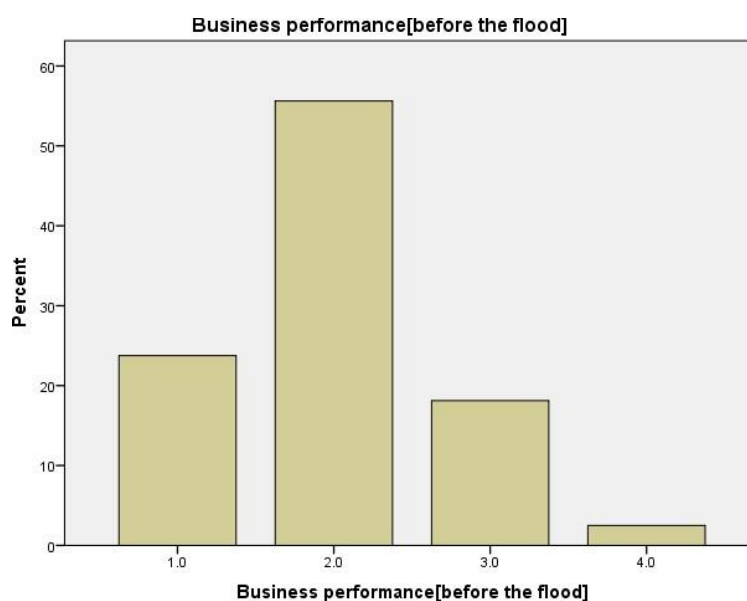


CHART-17

BUSINESS PERFORMANCE (BEFORE THE FLOOD)

TABLE – 18

S.NO	PARTICULARS	NO.OF RESPONDENT	PERCENTAGE
1	Excellent	23	14.3
2	Growing	32	20
3	Struggling	75	46.9
4	Loss	30	18.8
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION

The table-18 represents that 14.3% of the respondents experienced excellent business performance after the flood, 20% of the respondents experienced growing business performance after the flood, 46.9% of the respondents experienced struggling business performance after the flood, 18.8% of the respondents experienced loss in business performance after the flood.

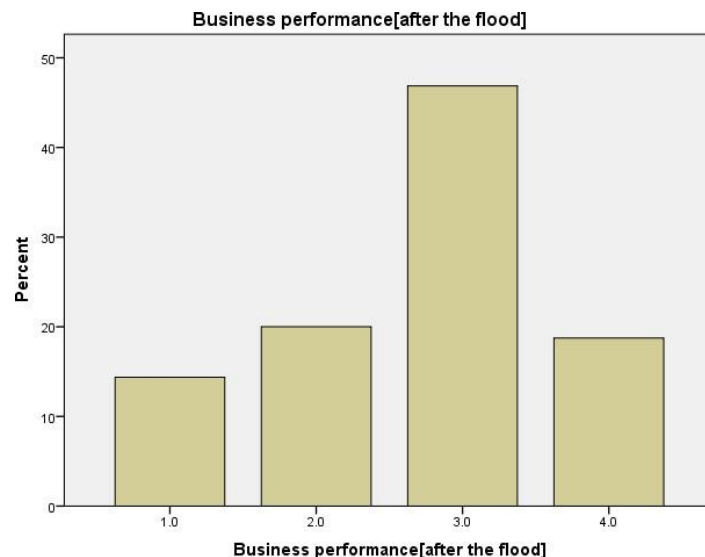


CHART-18

MIGRATION LEVEL (BEFORE THE FLOOD)

TABLE-19

S.NO	PARTICULARS	NO.OF RESPONDENTS	PERCENTAGE
1	High	26	16.2
2	Moderate	93	58.2
3	Low	26	16.2
4	None	15	9.4
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION

The table-19 represents that 16.2% of the respondents experienced high migration level before the flood, 58.2% of the respondents experienced moderate migration level before the flood, 16.2% of the respondents experienced low migration level before the flood, 9.4% of the respondents did not experience any migration before the flood.

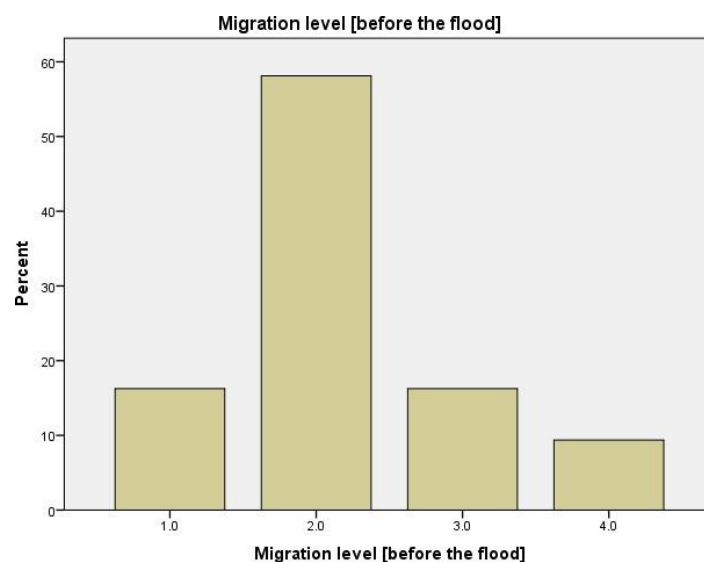


CHART-19

MIGRATION LEVEL (AFTER THE FLOOD)

TABLE-20

S.NO	PARTICULARS	NO. OF RESPONDENTS	PERCENTAGE
1	High	60	37.5
2	Moderate	36	22.5
3	Low	52	32.5
4	None	12	7.5
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION

The table-20 represents that 37.5% of the respondents experienced high migration level after the flood, 22.5% of the respondents experienced moderate migration level after the flood, 32.5% of the respondents experienced low migration level after the flood, 7.5% of the respondents who doesn't experience any migration after the flood. So there is a a lot of migration after the flood.

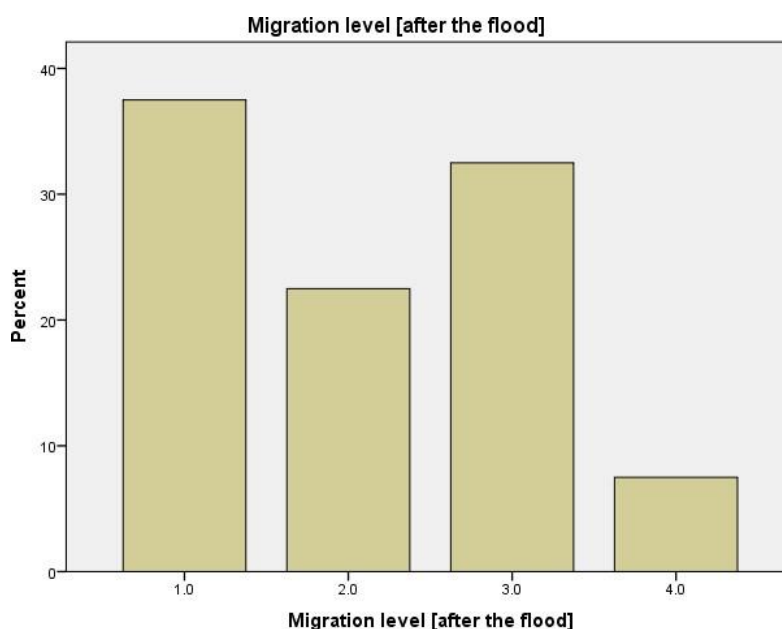


CHART-20

PROPERTY VALUES (BEFORE THE FLOOD)

TABLE-21

S.NO	PARTICULARS	NO. OF RESPONDENTS	PERCENTAGE
1	High	54	33.7
2	Moderate	95	59.4
3	Low	11	6.9
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION

The table-21 represents that 33.7% of the respondents thinks values of the property are high before the flood, 59.4% of the respondents thinks values of the property are moderate before the flood, 6.9% of the respondents thinks values of the property are low before the flood.

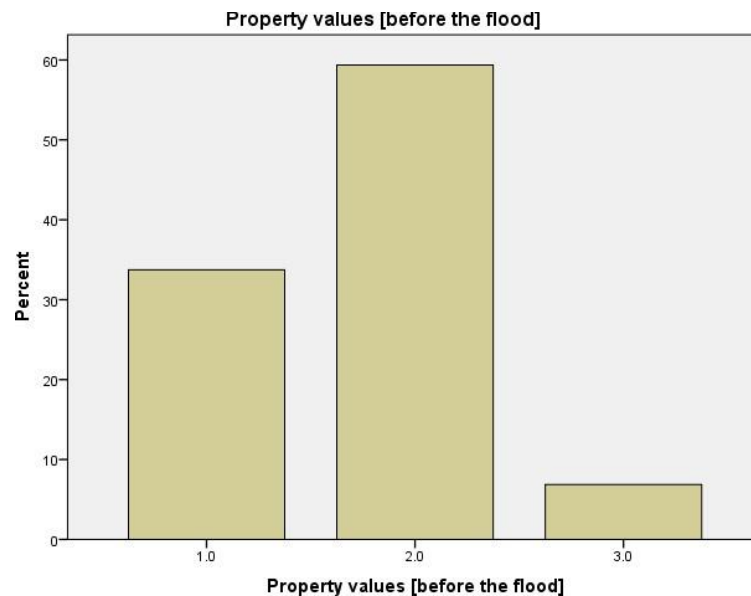


CHART-21

PROPERTY VALUES (AFTER THE FLOOD)

TABLE-22

S.NO	PARTICULAR	NO.OF RESPONDENTS	PERCENTAGE
1	High	40	25
2	Moderate	41	25.6
3	Low	79	49.4
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION

The table-22 represents that 25% of the respondents thinks values of the property are high after the flood, 25.6% of the respondents thinks values of the property are moderate after the flood, 49.4% of the respondents thinks values of the property are low after the flood.

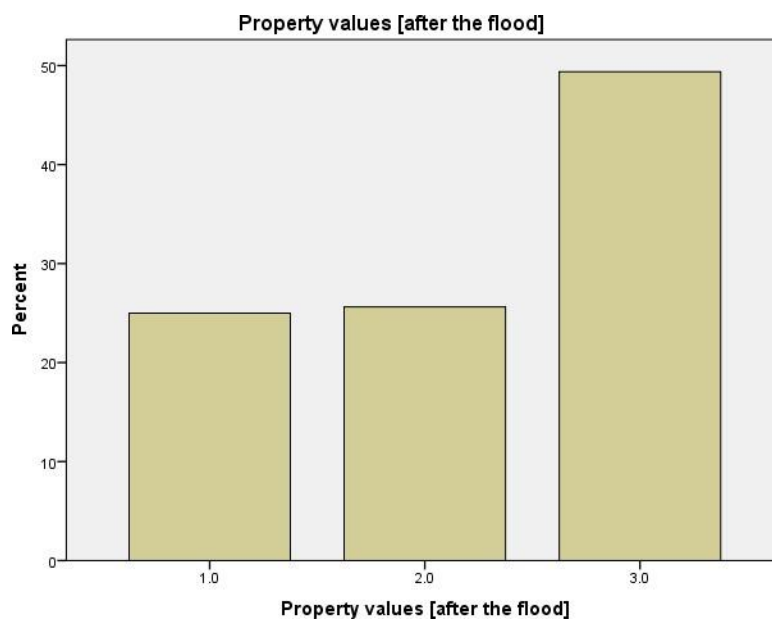


CHART-22

KNOWLEDGE ABOUT THE FLOOD (BEFORE THE FLOOD)

TABLE-23

S.NO	PARTICULARS	NO.OF RESPONDENTS	PERCENTAGE
1	Excellent	34	21.2
2	Somewhat	110	68.8
3	Neutral	14	8.8
4	None	2	1.2
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION

The table-21 represents that 21.2% of the respondents had excellent knowledge about floods before the flood, 68.8% of the respondents had somewhat knowledge about floods before the flood, 1.2% of the respondents had neutral knowledge about floods before the flood, 1.2% of the respondents had no knowledge about floods before the flood.

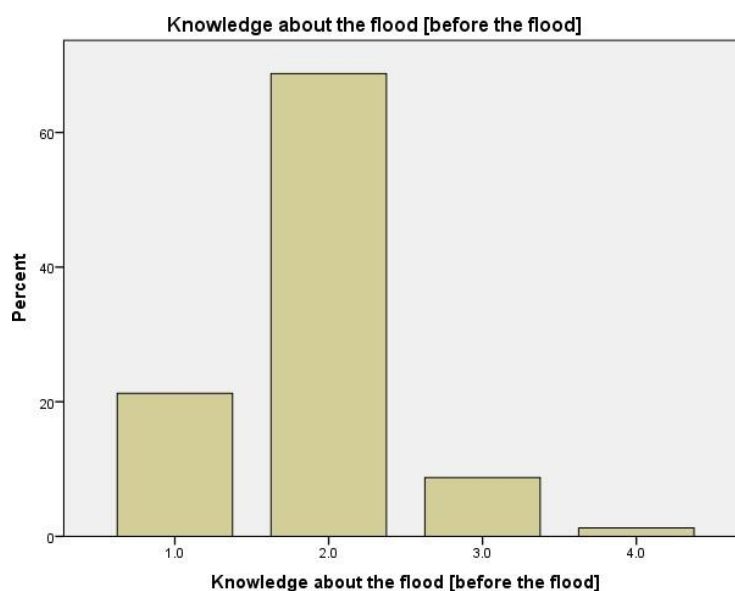


CHART-23

KNOWLEDGE ABOUT THE FLOOD (AFTER THE FLOOD)

TABLE-24

S.NO	PARTICULAR	NO. OF RESPONDENTS	PERCENTAGE
1	Excellent	81	50.6
2	Somewhat	36	22.5
3	Neutral	34	21.3
4	None	9	5.6
	TOTAL	160	100

SOURCE :Primary data

INTERPRETATION

The table-22 represents that 50.6% of the respondents had excellent knowledge about floods after the flood, 22.5% of the respondents had somewhat knowledge about floods after the flood, 21.3% of the respondents had neutral knowledge about floods after the flood, 5.6% of the respondents had no knowledge about floods after the flood.

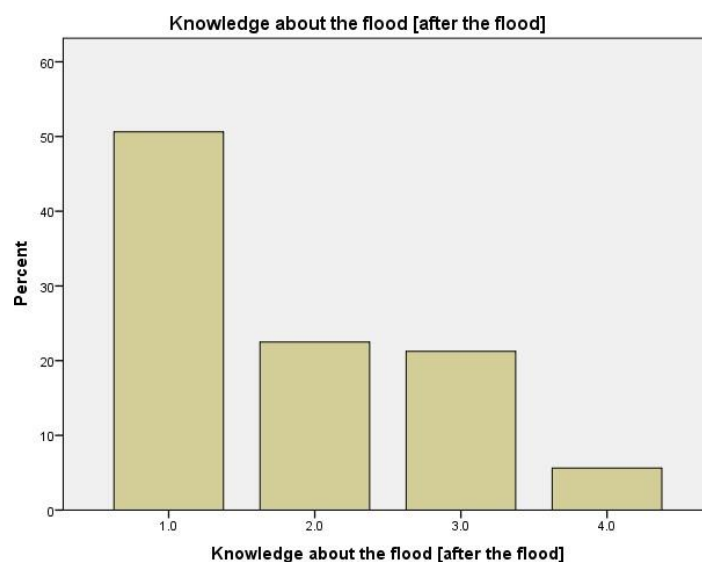


CHART-24

IMPACT ON LONG TERM RECOVERY

TABLE-25

S.NO	PARTICULAR	NO. OF RESPONDENTS	PERCENTAGE
1	Fully recovered	73	45.6
2	Partially recovered	72	45
3	Still recovered	15	9.4
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-25 represents that 45.6% of the respondents fully recovered from the flood impact, 45% of the respondents partially recovered from the flood impact, 9.4% of the respondents still recovering from the impact.

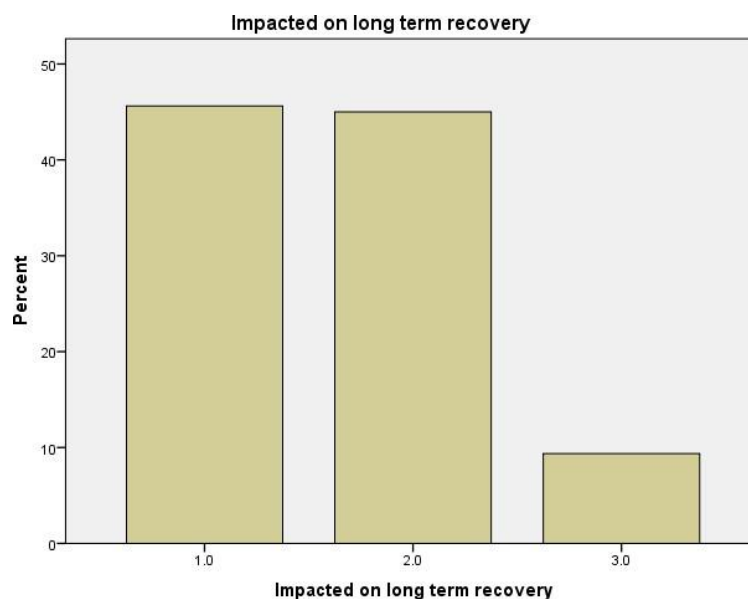


CHART-25

DAYS TAKEN TO RECOVER

TABLE-26

S.NO	PARTICULAR	NO. OF RESPONDENTS	PERCENTAGE
1	One month	42	26.3
2	Two months	76	47.5
3	Three months	42	26.2
	TOTAL	160	100

SOURCE: Primary data INTERPRETATION:

The table-26 represents that 26.3% of the respondents recovered from the flood impact within one month, 47.5% of the respondents recovered from the flood impact within two months, 26.2% of the respondents recovered from the flood impact within three months.

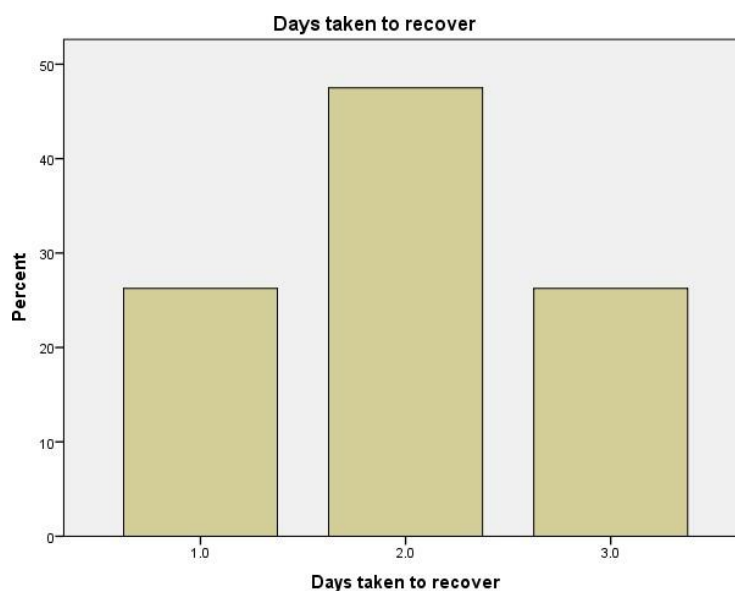


CHART- 26

LONG TERM CONSEQUENCES OF THE FLOOD

TABLE-27

S.NO	PARTICULARS	NO. OF RESPONDENTS	PERCENTAGE
1	Economic loss	53	33.1
2	Environmental loss	71	44.4
3	Community displacement	36	22.5
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-27 represents that 33.1% of the respondent's long-term consequences of the flood is economic loss, 44.4% of the respondent's long term consequences of the flood is environmental loss, 22.5% of the respondent's long term consequence of the flood is community displacement.

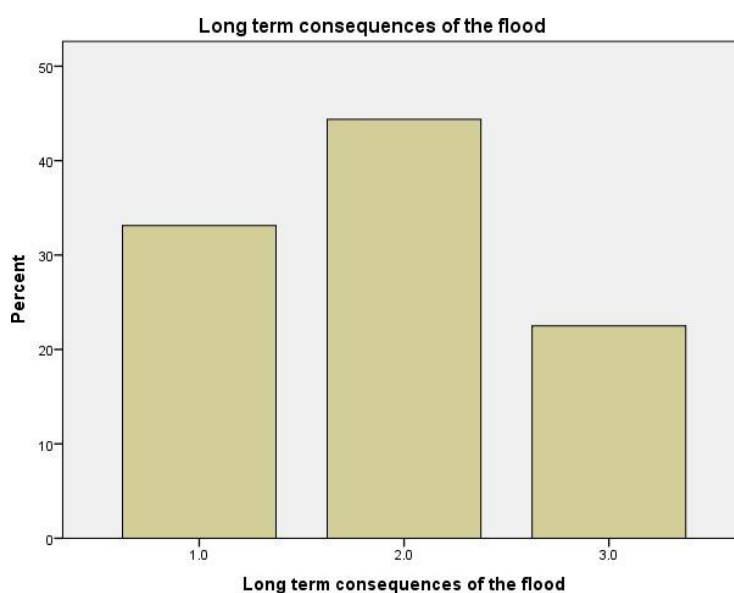


CHART-27

FLOOD WARNINGS OR INFORMATION BEFORE THE EVENT

TABLE-28

S.NO	PARTICULARS	NO. OF RESPONDENTS	PERCENTAGE
1	Official government alert	32	20
2	Local media	70	43.7
3	Social media	34	21.3
4	Community network	24	15
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-28 represents that 20% of the respondents received flood warning through official government alert, 43.7% of the respondents received flood warning through local media, 21.3% of the respondents received flood warning through social media, 15% of the respondents received flood warning through community network.

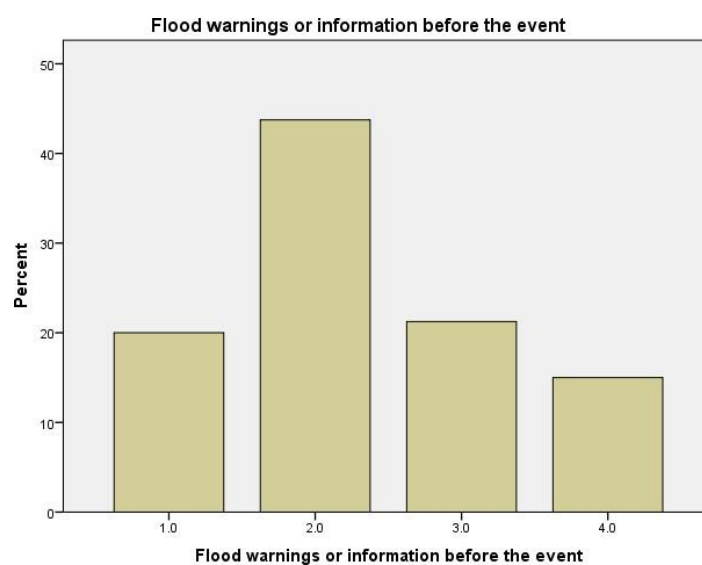


CHART-28

EARLY WARNING SYSTEM BY THE GOVERNMENT

TABLE-29

S.NO	PARTICULARS	NO. OF RESPONDENTS	PERCENTAGE
1	Very satisfied	50	31.2
2	Satisfied	57	35.7
3	Neutral	17	10.7
4	Dissatisfied	34	21.2
5	Very dissatisfied	2	1.2
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-29 represents that 31.2% of the respondents are very satisfied with early warning system by the government, 35.7% of the respondents are satisfied with early warning system by the government, 10.7% of the respondents is neither satisfied nor dissatisfied, 21.2% of the respondents are dissatisfied with early warning system by the government, 1.2% of the respondents are very dissatisfied with early warning system by the government.

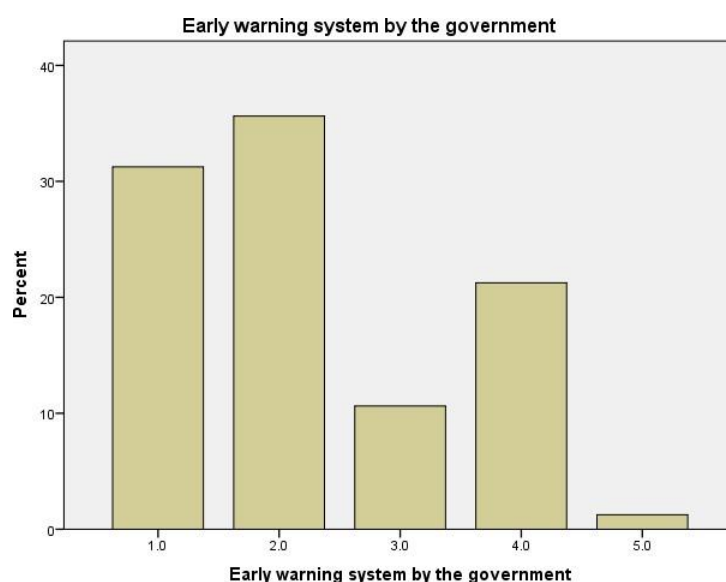


CHART-29

EMERGENCY SERVICE DURING THE FLOOD

TABLE-30

S.NO	PARTICULARS	NO. OF RESPONDENTS	PERCENTAGE
1	Very satisfied	28	17.5
2	Satisfied	66	41.2
3	Neutral	32	20
4	Dissatisfied	30	18.8
5	Very dissatisfied	4	2.5
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-30 represents that 17.5% of the respondents are very satisfied with emergency service by the government, 41.2% of the respondents are satisfied with emergency service by the government, 20% of the respondents is neither satisfied nor dissatisfied, 18.8% of the respondents are dissatisfied with emergency service by the government, 2.5% of the respondents are very dissatisfied with emergency service by the government.

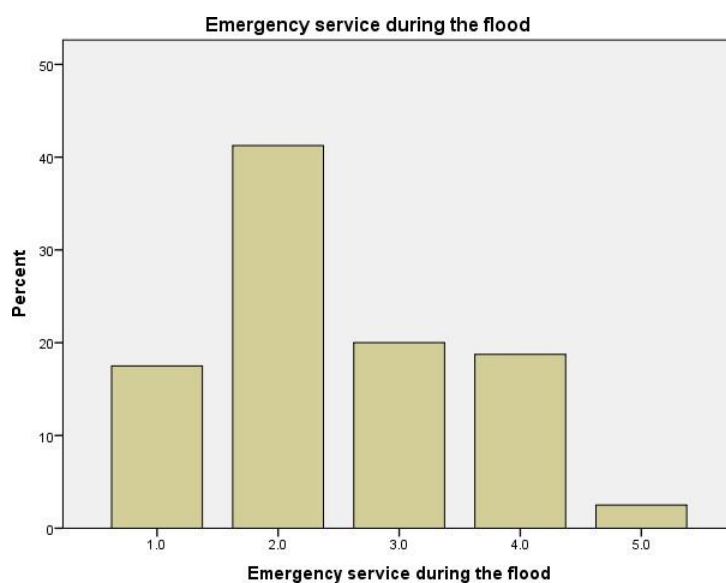


CHART-30

ASSISTANCE FROM THE GOVERNMENT DURING THE FLOOD

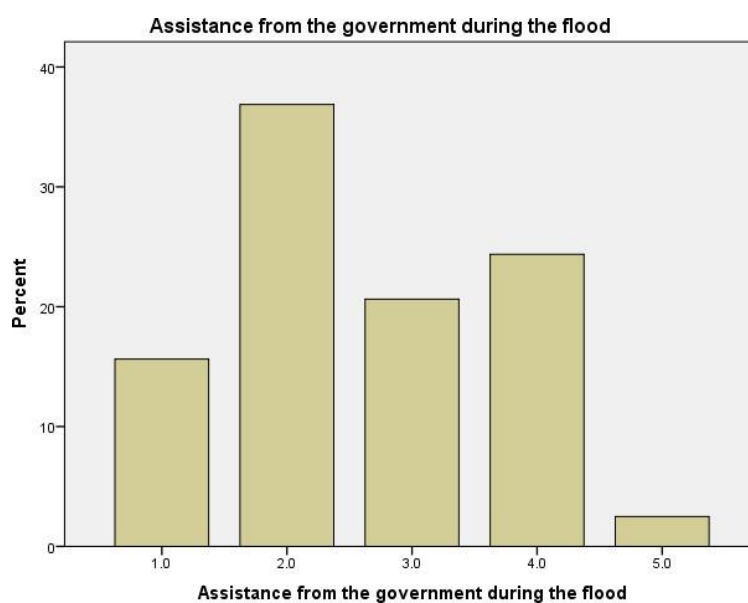
TABLE-31

S.NO	PARTICULARS	NO. OF RESPONDENTS	PERCENTAGE
1	Very satisfied	25	15.6
2	Satisfied	59	36.9
3	Neutral	33	20.6
4	Dissatisfied	39	24.4
5	Very dissatisfied	4	2.5
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-31 represents that 15.6% of the respondents are very satisfied with assistance from the government during the flood, 36.9% of the respondents are satisfied with assistance from the government during the flood, 20.6% of the respondents is neither satisfied nor dissatisfied, 24.4% of the respondents are dissatisfied with assistance from the government during the flood, 2.5% of the respondents are very dissatisfied with assistance from the government during the flood.



\CHART-31

RECOVERY ACTIVITIES FROM THE GOVERNMENT

TABLE-32

S.NO	PARTICULARS	NO. OF RESPONDENTS	PERCENTAGE
1	Very satisfied	29	18.1
2	Satisfied	60	37.5
3	Neutral	32	20
4	Dissatisfied	35	21.9
5	Very dissatisfied	4	2.5
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-32 represents that 18.1% of the respondents are very satisfied with recovery activities from the government during the flood, 37.5% of the respondents are satisfied with recovery activities from the government during the flood, 20% of the respondents is neither satisfied nor dissatisfied, 21.9% of the respondents are dissatisfied with recovery activities from the government during the flood, 2.5% of the respondents are very dissatisfied with recovery activities from the government during the flood.

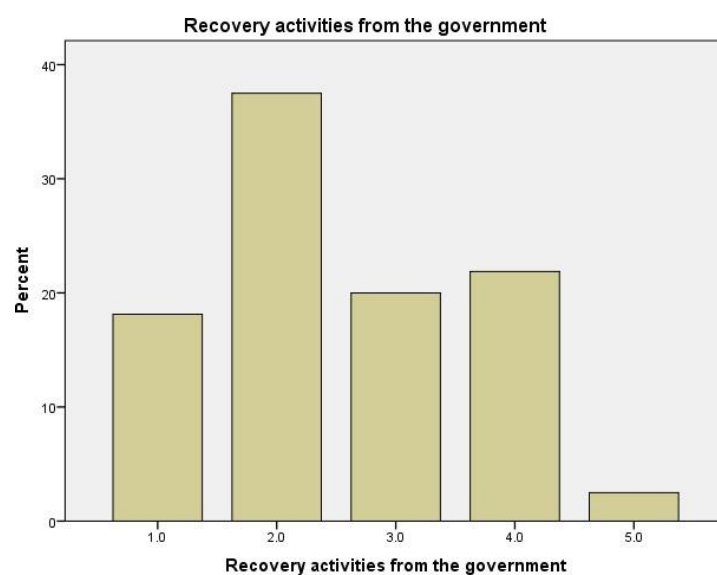


CHART-32

COMPENSATION FROM THE GOVERNMENT

TABLE-33

S.NO	PARTICULARS	NO. OF RESPONDENTS	PERCENTAGE
1	Very satisfied	20	12.5
2	Satisfied	59	36.9
3	Neutral	26	16.2
4	Dissatisfied	44	27.5
5	Very dissatisfied	11	6.9
	TOTAL	160	100

SOURCE: Primary data

INTERPRETATION:

The table-33 represents that 12.5% of the respondents are very satisfied with compensation from the government during the flood, 36.9% of the respondents are satisfied with compensation from the government during the flood, 16.2% of the respondents is neither satisfied nor dissatisfied, 27.5% of the respondents are dissatisfied with compensation from the government during the flood, 6.9% of the respondents are very dissatisfied with compensation from the government during the flood.

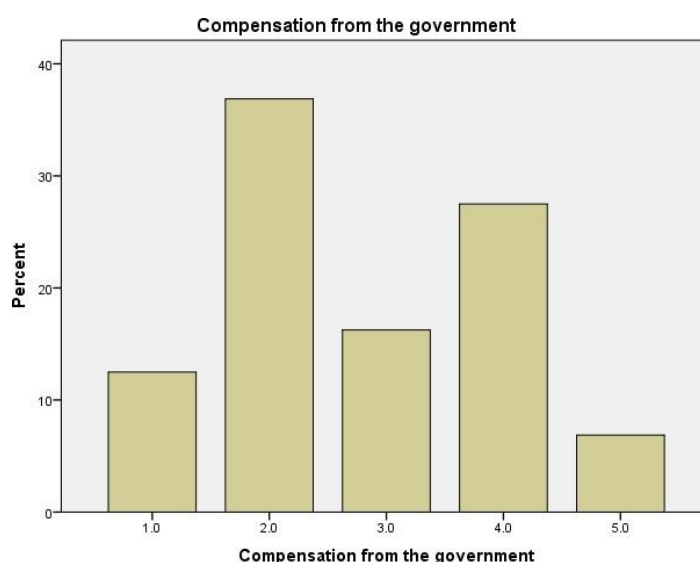


CHART-33

CHI SQUARE TEST

Chi square test is also known as goodness fit test is a statistical test used to compare observed result with the expected value. This test is used to determine if a difference between the observed data and expected data is due to chance or if it is due to a relationship between the variable that the study has taken.

CHI – SQUARE ANALYSIS:

Independent factor:

The independent factor here seems to be the damages experienced by the respondent.
(e.g., house damage, vehicle damage, land and agriculture damage)

Dependent factor:

The dependent factor here seems to be the respondent satisfaction level towards the government compensation.(e.g., satisfaction ratings related to compensation received)

Hypothesis:

✚ Null Hypothesis(H0):

There is no association between damages experienced by the respondents and their satisfaction level from the government's compensation.

✚ Alternative Hypothesis(H1):

There is association between damages experienced by the respondents and their satisfaction level from the government's compensation.

CALCULATION:

Damages experienced by the respondent* respondent's satisfaction level towards government compensation

(OBSERVED VALUE)

Observed value	Very satisfied	Satisfied	Neutral	Dissatisfied	Very dissatisfied	TOTAL
House damage	20	20	0	0	0	40
Vehicle damage	0	39	26	4	0	69
Land damage	0	0	0	33	0	33
Agriculture	0	0	0	7	11	18
TOTAL	20	59	26	44	11	160

(EXPECTED VALUE)

Observed value	Very satisfied	Satisfied	Neutral	Dissatisfied	Very dissatisfied	TOTAL
House damage	5.0	14.8	6.5	11.0	2.8	40.0
Vehicle damage	8.6	25.4	11.2	19.0	4.7	69.0
Land damage	4.1	12.2	5.4	9.1	2.3	33.0
Agriculture	2.3	6.6	2.9	5.0	1.2	18.0
TOTAL	20.0	59.0	26.0	44.0	11.0	160.0

P Value= 0.000

Interpretation:

- Pearson chi-square value: 295.707
- Degree of freedom(df): 12
- Asymptotic sig(2 sided): 0.000

Since the p-value (0.000) is much lower than the typical significance level of 0.05, we reject the null hypothesis. This indicates that there is association between damages experienced by the respondents and their satisfaction level from the government's compensation.

The result indicates that damages have a statistically significant effect on the compensation distribution, with the chi-square test confirming the strong relationship between these variables.

ANOVA:

Independent factor:

The independent factor here seems to be the damages experienced by the respondent. (e.g., house damage, vehicle damage, land and agriculture damage)

Dependent factor:

The dependent factor here seems to be the respondent satisfaction level towards the government compensation.(e.g., satisfaction ratings related to compensation received)

Hypothesis:

✚ **Null Hypothesis(H0):**

There is no association between damages experienced by the respondents and their satisfaction level from the government's compensation.

✚ **Alternative Hypothesis(H1):**

There is association between damages experienced by the respondents and their satisfaction level from the government's compensation.

Level of significance:

The typical level of significance is 0.05(5%). This is not explicitly stated in the output, but unless otherwise mentioned, this is the standard.

ANOVA

Sources of variation	Sum of squares	Df	Mean square	F	Sig.
Between groups	178.430	3	59.477	222.165	.000
Within groups	41.763	156	.268		
Total	220.194	159			

Anova output:

F- statistics: 222.165

p- value(sig.):0.000

Interpretation:

The p-value is 0.000, which is lower than the typical significance level of 0.05.

Conclusion: since the p-value is lower than 0.05, we can reject null hypothesis. This means there is There is association between damages experienced by the respondents and their satisfaction level from the government's compensation.

The ANOVA test result indicate that There is association between damages experienced by the respondents and their satisfaction level from the government's compensation.

CHAPTER V

FINDINGS

Findings are basically the key outcome of the research. It is basically key facts discovered during research. Here findings do not always mean factual information because conductive research relies on results and implications rather than measurable facts.

BASED ON PERCENTAGE ANALYSIS:

- ✚ Among 160 respondents, 65.6% of the respondents are female.
- ✚ Out of all the 160 respondents, 56.9% of the respondents are 20-29 years old.
- ✚ 32.5% of the respondents are from Gandhi nagar.
- ✚ 41.9% of the respondents are extremely affected by the flood.
- ✚ Most of the respondents 43.1% experienced loss of belongings.
- ✚ Among 160 respondents, 43.1% of the respondents experienced vehicle damage.
- ✚ Most of the respondents 34.4% displaced from their home for three weeks.
- ✚ 45% of the respondents experienced the weeks of electricity down during the flood.
- ✚ Most of the respondents 43.1% household income were affected by losing their job because of flood.
- ✚ 54.3% of the respondents faced significant impact on their daily life routine.
- ✚ 50% of the respondents significantly suffered for food during the flood.
- ✚ 56.9% of the respondents significantly faced health issues due to the flood.
- ✚ 53.1% of the respondents significantly faced emotional impact due to the flood.
- ✚ 50.6% of the respondent social interaction has been significantly affected.
- ✚ 50% of the respondent children's education has been significantly affected.
- ✚ Out of all 160 respondents, 55.6% of the respondent's business performance is in the stage of growing before the flood.
- ✚ Out of all 160, 46.9% of the respondent's business performance is in the stage of struggling after the flood.
- ✚ Most of the respondents 58.2% thinks there is a moderate migration level in their area before the flood.
- ✚ Most of the respondents 37.5% thinks there is a high migration level in their area after the flood.
- ✚ 59.4% of the respondents says property values are moderate before the flood.
- ✚ 49.4% of the respondents says property values are low after the flood.

- ✚ Most of the respondents 68.8% had somewhat knowledge about flood before the event.
- ✚ Most of the respondents 50.6% have excellent knowledge about flood after the flood.
- ✚ Out of 160 respondents 45% partially recovered from the flood impact.
- ✚ Among 160 respondents, for 47.5% it takes two months to recover from the flood.
- ✚ 44.4% of the respondent's long-term consequence of the flood is environmental loss.
- ✚ Most of the respondents 43.7% received flood warning from local social media.
- ✚ 35.7% of the respondents are satisfied with the early warning system by the government.
- ✚ 41.2% of the respondents are satisfied with emergency service of the government during the flood.
- ✚ 36.9% of the respondents are satisfied with the assistance from the government during the flood.
- ✚ 37.5% of the respondents are satisfied with recovery activities from the government.
- ✚ 36.9% of the respondents are satisfied with government compensation.

BASED ON TEST ANALYSIS:

There is a association between damages experienced by the respondents and their satisfaction level towards government's compensation.

The anova test results indicates that there is a association between damages and government's compensation.

SUGGESTION:

- Government should analysis the damages experienced by the people and give the compensation accordingly.
- Government's compensation process needs improvement in terms of transparency, consistency, and efficiency.
- As most of the respondent suffered for food during flood government should consider helicopter arrangements during flood to avoid such situation.
- Most of the respondent received flood warning through social media so government should improve their early warning system.

CHAPTER VI

CONCLUSION:

The study reveals a significant association between the damages experienced by people during the flood and the compensation provided by the government. It highlights that individual who suffered greater losses, such as damage to property, livelihood and personal belongings were more likely to receive higher compensation. Addressing this issue would ensure fair relief for all affected individuals, helping them recover faster and build resilience against future disaster. The findings emphasis the need for reforms in disaster management policies to prioritize the need of the most vulnerable.

BIBLIOGRAPHY:

REFERENCE

1. Singh, R. (2015), "Historical Analysis of Flood Events in South India"
2. Kumar, A. & Rao, S. (2016), "Vulnerability Assessment of Coastal Communities in Tamil Nadu"
3. Sharma, P. (2017), "Flooding and Its Impact on Agricultural Output in Coastal Areas"
4. Menon, S. et al. (2018), "Health Risks Post-Flooding in Tamil Nadu"
5. Gupta, N. & Verma, T. (2019), "Economic Impact of Natural Disasters: Case Study of Tuticorin."
6. Chatterjee, R. (2020), "Displacement Patterns Due to Floods in Coastal Tamil Nadu"
7. Rani, L. & Subramanian, K. (2021), "Flood Response Strategies: Government and NGO Initiatives in Tuticorin"
8. Joseph, A. (2022), "Building Resilience in Flood-Prone Communities"
9. Das, M. & Reddy, V. (2023), "Infrastructure Damage Assessment Post-Floods in Tuticorin"
10. Patel, S. (2023), "Gender Dimensions of Flood Vulnerability"
11. Iyer, N. & Narayan, P. (2023), "Economic Recovery Strategies in Flood-Affected Regions"
12. Rao, K. (2023), "Cultural Impact of Flood Disasters in Coastal Communities"
13. Srinivasan, R. (2023), "Climate Change and Flood Risk: Insights from Tamil Nadu"
14. Kaur, D. (2024), "The Role of Indigenous Knowledge in Disaster Preparedness"
15. Nambiar, S. (2024), "Policy Responses to Flood Management in Tamil Nadu"

ANNEXURE

1. Gender

- Male
- female

2. What is your age?

- a)Below 20
- b)20 – 29
- c)30 – 39
- d)Above 40

3. Where is your place?

- a)Thiresh nagar
- b)JJ nagar
- c)Gandhi nagar
- d)Muthaiya puram

4. How severely were you affected by the flood?

- a)Extremely
- b)Significantly
- c)Moderately
- d)slightly

5. Indicate the specific impacts experienced due to the flood.

- a)Damage to property
- b)Loss of belongings
- c)Health issues
- d)loss of income or livelihood

6. Approximately how many people were affected by the flood in your area?

- a)below 300
- b)around 500
- c)around 800
- d)more than 1000

7. what is your major property damage you experienced due to the flood?

- House damage
- vehicles damage
- land damage
- agriculture

8. how long were you displaced from a home due to the flood?

- less than a week
- 2-3 weeks
- more than 3 weeks
- not at all

8. How many days the electricity were down during the flood?

- 3-4 days
- one week
- two weeks
- around a month

9. how did the flood affected your household income?

- reduced income.
- loss of job
- business interruption

10. How severely these impacts affected you?

	EXTREMELY	SIGNIFICANTLY	MODERATELY	SLIGHTLY
impact on your daily life routine				
How much did you suffer for food				
Health issues faced during the flood				
Emotional impact of the flood				
Affected on social interactions				
Affected on children's education				

17. How is your business performance?

	Excellent	Growing	Struggling	loss
Before the flood				
After the flood				

18. How is migration level in your area?

	High	Moderate	Low	None
Before the flood				
After the flood				

19. property values

	High	moderate	low
Before the flood			
After the flood			

20. How is your knowledge about the flood?

	Excellent	Somewhat	Neutral	None
Before the flood				
After the flood				

21..How has the flood event impacted your long-term recovery?

- fully recovered
- partially recovered
- still recovering

22. how many days it takes your area to recover from the impact?

- one month
- two months
- three months

23. what are the long-term consequences of the flood in your area?

- Economic losses
- environmental damage
- community displacement

24..How did you received flood warnings or information before the event?

- official government alerts
- local media
- social media
- community networks

25. How satisfied are you with these government activities?

	Very satisfied	Satisfied	Neutral	Dissatisfied	Very dissatisfied
Early warning system by the government					
Emergency service during the flood					
Assistance from the government					
during the flood					
Recovery activities from the local authorities					
Compensation from the government					